How Literacy and Age at Marriage Shape Family Size in Portugal, 1979*

A Statistic Research Using Poisson Regression Models

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*Code and data are available at: https://github.com/Jie-jiao05/Family_Size.

1 Introduction

1.1 Why this topic is important

1.2 3 citation relate with our research

1.3 Our objective, goal, how the citation related to, what we gonna do

1.4 Brief introduction of our research

2 Method

2.1 Use GLM, possion, (reason), what is x how to interoperate

3 Result

3.1 Data Description

This dataset originates from the 1979-80 Portuguese Fertility Survey, conducted as part of the World Fertility Survey (WFS) program and archived by Princeton University's Office of Population Research (OPR).

The Dataset includes 5,148 observations and 8 variables, covering factors such as literacy, marriage age, pregnancies, children, and regional demographics. Table 1 it shows a summary information of our dataset.

Table 1: Dataset Summary

```
       X              age         ageMarried          monthsSinceM
 Min.   :   1   Min.   :15.00   Length:5148        Min.   :  0.0
 1st Qu.:1288   1st Qu.:28.75   Class :character   1st Qu.: 73.0
 Median :2574   Median :36.00   Mode  :character   Median :148.0
 Mean   :2574   Mean   :35.30                      Mean   :155.2
 3rd Qu.:3861   3rd Qu.:43.00                      3rd Qu.:230.0
 Max.   :5148   Max.   :49.00                      Max.   :431.0
  pregnancies        children          sons           literacy
 Min.   : 0.000   Min.   : 0.00   Min.   : 0.000   Min.   :0.0000
 1st Qu.: 1.000   1st Qu.: 1.00   1st Qu.: 0.000   1st Qu.:1.0000
 Median : 2.000   Median : 2.00   Median : 1.000   Median :1.0000
 Mean   : 2.282   Mean   : 2.26   Mean   : 1.187   Mean   :0.8871
 3rd Qu.: 3.000   3rd Qu.: 3.00   3rd Qu.: 2.000   3rd Qu.:1.0000
 Max.   :16.000   Max.   :17.00   Max.   :10.000   Max.   :1.0000
  region10.20k       region20k.      regionlisbon     regionlt10k
 Min.   :0.00000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.00000   Median :0.0000   Median :0.0000   Median :1.0000
 Mean   :0.08411   Mean   :0.1132   Mean   :0.0913   Mean   :0.6803
 3rd Qu.:0.00000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000
 Max.   :1.00000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
  regionporto
 Min.   :0.00000
 1st Qu.:0.00000
 Median :0.00000
 Mean   :0.03108
 3rd Qu.:0.00000
 Max.   :1.00000
```

3.2 Set up model

3.3 Model selection (use model with offset)

3.4 Use likelyhood rario test to do the performance , to see the efficiency between model

3.5 Test overdispersion

4 Conclusion

Appendix

5 References